

Thomas Falconer

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Education

Technical University of Denmark <i>Ph.D. in Machine Learning and Economics</i>	June 2022 – June 2025 <i>Kgs. Lyngby, Denmark</i>
University College London <i>M.Sc. in Data Science (Distinction, Top in Year)</i>	Sept 2019 – Nov 2020 <i>London, United Kingdom</i>
University of Amsterdam <i>B.Sc. in Operations Research (Erasmus Exchange)</i>	Sept 2016 – June 2017 <i>Amsterdam, Netherlands</i>
Heriot-Watt University <i>B.Eng. in Engineering (First Class, Top in Year)</i>	Sept 2014 – June 2019 <i>Edinburgh, United Kingdom</i>

Experience

Quantitative Researcher (Algorithmic Trading) <i>InCommodities</i>	June 2025 – present <i>Aarhus, Denmark</i>
- Develop probabilistic ML models to forecast short-term power price movements in Europe. - Generate signals and execution policies for continuous trading, navigating OB dynamics, liquidity, and forecast updates. - Apply stochastic programming over forecast distributions to optimize bidding in day-ahead and intraday auctions. - Conduct rigorous backtesting and statistical validation to assess model reliability and strategy performance. - Build real-time dashboards to monitor algorithms, data quality, and model drift. - Implement CI/CD pipelines for automated model retraining, validation, and production deployment.	
Machine Learning Engineer (Contract) <i>Freelance (During Ph.D.)</i>	June 2022 – June 2022 <i>Copenhagen, Denmark</i>
- Designed and deployed end-to-end ML solutions for clients, including data ingestion, feature engineering, model development, and performance monitoring. - Developed maintainable codebases with testing, packaging, and documentation.	
Data Scientist (Algorithmic Trading) <i>Arenko</i>	Nov 2020 – June 2022 <i>London, United Kingdom</i>
- Built and maintained flexibility trading algorithms for renewable and battery assets, using ML and constrained optimization to maximize revenue across energy markets and ancillary services. - Implemented ensembles of forecasting models, including gaussian processes, neural networks and gradient boosting, plus simpler statistical models. - Created interactive visualisations of market opportunities (matplotlib, plotly, Dash, Streamlit). - Built MLOps pipelines in AWS, including feature store, model versioning (MLflow), low-latency model serving (FastAPI), data engineering and orchestration (Prefect) and digestion (RDBMS) in a microservice framework. - Created interactive visualisations of market opportunities (matplotlib, Plotly, Dash, Streamlit). - Mentored junior data scientists and helped grow the data science team.	
Machine Learning Researcher (Intern) <i>Invenia Labs</i>	Apr 2019 – Sept 2019 <i>London, United Kingdom</i>
- Developed ML pipelines to forecast nodal price spreads for virtual/convergence bidding across US ISO markets. - Applied graph neural networks to model grid topology for congestion forecasting and optimal power flow. - Co-authored NeurIPS workshop paper and IEEE journal publication on ML for power systems.	

Technical Skills

Analysis: pandas, Polars, NumPy; experiment design, statistical testing, feature engineering, model explainability
Modelling: scikit-learn, LightGBM, PyTorch, TensorFlow; Bayesian inference, uncertainty quantification, validation
Optimization: Pyomo, CVXPY, Gurobi; stochastic programming, constrained optimization, DRO, metaheuristics
MLOps: Docker, FastAPI, MLflow, Prefect, Argo; ETL, versioning, model serving, orchestration, monitoring
Infrastructure: AWS (S3, EC2, Lambda), PostgreSQL, CI/CD, logging (Humio), package distribution (Gemfury)
Visualization: matplotlib, Plotly, Dash, Streamlit